Dmitri Manajev

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Summary

Robotics and Reinforcement Learning Engineer with 15+ years of software engineering experience, including software architecture, engineering leadership, technical product management, and entrepreneurship. Currently developing and commissioning ROS 2 software in C++ and Python for autonomous robotic systems, spanning navigation, motion control, systems integration, simulation, imitation learning, and real-robot deployment. Earlier work includes regulated medical software, cloud infrastructure, mobile and web products, and distributed systems.

Skills

- **Robotics Software**: ROS 2 and ROS 1, ros2_control, Nav2, Behavior Trees, MoveIt 2 / MoveIt Pro, tf2, URDF/Xacro; navigation, motion planning and control, manipulation, systems integration, failure recovery, commissioning and hardware–software debugging
- **Robot Learning & Reinforcement Learning**: PPO, SAC, TD3, DDPG; multi-agent RL (MAPPO, HAPPO, MADDPG); offline-to-online RL (RLPD); LeRobot policies (ACT, SmolVLA); reproducible training/evaluation, sim-to-real and real-robot deployment
- **AI/ML & Simulation**: PyTorch (primary), JAX, TensorFlow, NumPy, OpenCV; Gazebo, MuJoCo, Isaac Gym, Unity ML-Agents; Gymnasium, PettingZoo, SMACv2, Weights & Biases, Hugging Face Hub
- **Software Engineering & Infrastructure**: Linux (Ubuntu), Git, Docker, CMake, automated testing, CI/CD (GitHub Actions, Jenkins), Ansible, AWS and nginx
- **Programming Languages**: C++ and Python; additional production experience with JavaScript/TypeScript, Swift/Objective-C, Solidity, C#, and SQL/NoSQL
- **Architecture, Product & Leadership**: Software architecture, engineering-team leadership, technical product management, PRDs/specifications, roadmaps/backlogs, Agile delivery and regulated software releases

Languages

English (fluent), German (A2/B1)

Experience

- **Hivebotics** — autonomous robotics — Remote + on-site, Singapore
  Robotics Engineer (Independent Contractor) — *May 2026 – Present*
  - Develop and commission production ROS 2 software in C++ and Python for autonomous mobile cleaning robots, covering Nav2 navigation, MoveIt Pro manipulation, motion control and systems integration.
  - Improve real-world reliability through cross-stack debugging, recovery handling, simulation/real-robot validation and field diagnostics.
  - Supported on-site integration, testing and commissioning in Singapore.
- **Manajev AI Labs** — Zurich, CH
  Freelance Robotics / AI Software Engineer — *Mar 2026 – Present*
  - **Analytics company (Switzerland) — robotics simulation research** (Mar – May 2026): developed PX4/ROS 2 control and simulation prototypes in Gazebo and Unity; established reproducible development and GPU simulation environments.
  - **The Construct — course author** (Mar – May 2026): designed and built a physical-AI robot-arm course covering SO-101/ROS 2 integration, data collection, LeRobot/ACT policy training and edge deployment; public ROS 2 project and demos.
- **Independent Robotics & RL R&D** (self-directed) — Zurich, CH
  Self-directed robotics portfolio and structured RL study — *Dec 2024 – Feb 2026*
  - Built reproducible RL training/evaluation pipelines for multi-agent and continuous-control tasks (Unity ML-Agents, PettingZoo); ran algorithm benchmarks with documented experiment artifacts (GitHub).
  - Implemented ROS 2/C++ control loops for policy deployment; navigation, waypointing, manipulation and sim-to-real on differential-drive/holonomic bases and 5/6-DoF arms, incl. LeRobot imitation learning.

- **CryptoApps Manajev** — Zurich, CH
  Founder & Software Engineer — *Aug 2018 – Dec 2024*
  - Designed, built and shipped full-stack decentralized applications on Ethereum and Tron (Solidity, Node.js, Vue.js), incl. smart-contract security analysis and audits.
  - Built automated transaction-routing systems and execution bots for high-throughput distributed environments; optimized Ethereum/Tron/EOS node infrastructure for latency and cost.
- **Swissmeda AG** — Zurich, CH
  Product & IT Manager / Software Architect / DevOps — *Nov 2015 – Jun 2018*
  - Led the software engineering team (backlog planning, 1:1s, delivery); authored product specs/PRDs and drove iOS, desktop and backend releases under medical-software regulatory requirements.
  - Designed and ran AWS cloud infrastructure; migrated the company to GitHub, Jira, Confluence and Linux and standardized CI/CD and release workflows across mobile, desktop and backend.
  - Architected and shipped the **smop** iOS app for 3D implant plan review (**App Store**).
- **Swissmeda AG** — Zurich, CH
  Senior iOS Engineer — *Apr 2014 – Oct 2015*
  - Architected the multi-brand **smop** iOS app (3D implant plan viewing, slice navigation, case review); automated build/deploy with Fastlane and CocoaPods.
- **SpotMe (Shockfish SA)** — Lausanne, CH
  Senior Full-Stack Engineer — *Jan 2013 – Feb 2014*
  - Built real-time event features (voting, photo sharing, check-in, surveys, data viz) across backend and mobile web (Node.js, CouchDB, d3.js) for multiple client projects on tight deadlines.
- **Life Ideas** — St. Petersburg, RU
  Creator / Mobile Software Engineer — *Jan 2012 – Jan 2013*
  - Created **Instacam**, the first unofficial Instagram client for Windows Phone 7, with a photo-filter engine and 7 social-network integrations; **media coverage** and broad adoption.
- **Lanit-Tercom** — St. Petersburg, RU
  Intern to Software Engineer — *Jan 2010 – Dec 2012*
  - Developed mobile apps (iOS, WP7, Android), a project-management web app and a WPF/.NET desktop GUI for analytical instruments for outsourcing clients.

Education

- **St. Petersburg State University (SPbSU)** — St. Petersburg, Russia
  M.Sc.-equivalent Specialist Degree in Software Engineering & Mathematics — *2007 – 2012*
  Thesis: Collaborative recommendation system for tourist attractions.

Certifications & Training

- **Robotics Developer (Masterclass), The Construct** — *Sep 2025 – Mar 2026*
- **Humanoid Robotics Certificate, Unitree G1 (Barcelona), The Construct** — *Dec 2025*
- **Deep Reinforcement Learning (Nanodegree), Udacity** — *Nov 2024 – May 2025*
- **Data Analysis with Pandas & NumPy, Udacity** — *Sep 2024*
- **Machine Learning with PyTorch (Nanodegree), Udacity** — *2023*
- **Full Stack JavaScript Developer (Nanodegree), Udacity** — *2023*